

Integration by Parts Worksheet

Twenty problems covering polynomial products, logarithms, inverse trigonometric functions, definite integrals, and cyclic cases.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/worksheets/integration-by-parts/>

Name: _____ **Date:** _____

1. Evaluate $\int x e^x dx$.

2. Evaluate $\int x^2 e^x dx$.

3. Evaluate $\int x^3 e^x dx$.

4. Evaluate $\int x e^{2x} dx$.

5. Evaluate $\int x \sin x dx$.

6. Evaluate $\int x \cos x dx$.

7. Evaluate $\int x^2 \sin x dx$.

8. Evaluate $\int x^2 \cos x \, dx$.

9. Evaluate $\int \ln x \, dx$.

10. Evaluate $\int x \ln x \, dx$.

11. Evaluate $\int (\ln x)^2 \, dx$.

12. Evaluate $\int \arctan x \, dx$.

13. Evaluate $\int \arcsin x \, dx$.

14. Evaluate $\int_0^1 x e^x \, dx$.

15. Evaluate $\int_0^\pi x \sin x \, dx$.

16. Evaluate $\int e^x \cos x \, dx$.

17. Evaluate $\int e^x \sin x \, dx$.

18. Evaluate $\int x^4 e^x \, dx$.

19. Evaluate $\int x^2 \ln x \, dx$.

20. Evaluate $\int_1^e \ln x \, dx$.